Peng Li

Fluid and Thermal Modelling | CFD | Model Validation

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Professional Summary

Computational fluid dynamics researcher at KTH Royal Institute of Technology with industrial experience in conservation-based transient modelling, heat transfer, and coupled 0-D/3-D simulation. Develop Python models and numerical workflows to analyse pressure, temperature, and flow response; compare predictions with measurements and numerical reference solutions. Research spans compressible CFD, reduced combustion modelling, non-isothermal flows, and fluid–structure interaction (FSI). Seeking to contribute model development, scientific programming, and verification and validation skills to Blykalla's fluid-system analysis.

Technical Skills

Fluid and thermal modelling: Mass, momentum, and energy conservation; 0-D transient models; compressible flow; combustion and heat transfer; nozzle flow; non-isothermal low-Mach-number flows; natural convection.

Model development and coupling: Python-based physical models; coupled 0-D/3-D simulations; transient boundary conditions; finite-volume and finite-element methods; compressible LES; strong two-way CFD–FEM coupling.

Verification and validation: Comparison with measured pressure histories; parameter-sensitivity studies; mesh and time-step convergence; analytical/FEM cross-checks; reduced/detailed kinetic-model comparisons; assessment of model assumptions and limitations.

Programming and scientific computing: Python, MATLAB, C/C++, Fortran, Mathematica; scientific data analysis and visualisation; HPC simulation workflows using NAISS resources at PDC and NSC.

Simulation software: OpenFOAM, Simcenter STAR-CCM+, COMSOL Multiphysics, ABAQUS, ANSYS; Tecplot, Gnuplot, Inkscape, LaTeX.

Additional analysis: Acoustic wave and resonance modelling; structural eigenmodes; power and cross-spectra; coherence and phase; spectral proper orthogonal decomposition (SPOD); hyperelasticity and flow-induced vibration.

Selected Relevant Experience

Research Assistant – Transient Fluid and Thermal Modelling Dec 2018 – Mar 2020
Joyson Safety Systems, Aschaffenburg, Germany

- Developed Python-based 0-D models of pyrotechnical initiators using mass and energy conservation, combustion, heat transfer, and nozzle-flow relations to predict transient system behaviour.
- Generated pressure, temperature, mass-flow, and energy-flow boundary conditions and coupled initiator models to 3-D OpenFOAM simulations of shock-wave propagation in airbag inflators.
- Compared predicted pressure histories with closed-bomb measurements and investigated temperature and component sensitivities to assess performance and structural integrity.
- Connected detailed physical modelling to product-development questions; findings published in *Journal of Energy and Power Technology* following the M.Sc. thesis.

Ph.D. Researcher – Computational Fluid Dynamics and FSI Mar 2022 – Present
Department of Engineering Mechanics, FLOW, KTH Royal Institute of Technology, Stockholm

- Develop three-dimensional transient upper-airway aeroacoustic models coupling compressible LES with FEM-based structural mechanics in STAR-CCM+, including strong two-way FSI, hyperelastic materials, and moving overset meshes.
- Conduct mesh and time-step convergence studies and parameter studies of flow conditions, geometry, and material stiffness; analyse unsteady pressure, fluid loads, and structural response using spectral and modal methods.
- Verify an analytical coupled acoustic model against COMSOL FEM: frequency differences within 1.03% for the first ten soft-splitter common resonances in the tested geometry (unpublished manuscript).
- Run simulations on NAISS HPC resources at PDC and NSC; lead analysis, visualisation, and manuscript writing across fluid mechanics, structural dynamics, and acoustics, including a *Physics of Fluids* Featured Article.

Research Assistant – Reduced Combustion Modelling Sep 2020 – Aug 2021
Institute for Technical Thermodynamics, Karlsruhe Institute of Technology, Germany

- Applied the reaction–diffusion manifold (REDIM) method to premixed dimethyl-carbonate/air flame–wall interactions; validated reduced kinetic models by comparison with detailed kinetics.

Education

Ph.D. studies – Computational Fluid Dynamics Mar 2022 – Present
KTH Royal Institute of Technology, Stockholm, Sweden
Thesis: *Computational Aero-acoustics of Snoring with Impact on Obstructive Sleep Apnea Diagnosis*
Defence scheduled: 4 December 2026.
Supervision: Mihai Mihaescu, Marco Laudato, and Susann Boij.

M.Sc. in Mechanics (Mechanik) Oct 2016 – Jan 2020
Technical University of Darmstadt, Germany
Final grade: **1.33 (very good)**; thesis: **1.0** (German grading system).
Thesis: *Analysis and modeling of ballistic behaviour of pyrotechnical initiators and their impact in airbag inflators applications.*

B.Eng. in Engineering Mechanics Sep 2011 – Jun 2015
Dalian University of Technology, China
Overall mark: 86/100; thesis mark: 98/100.

Selected Peer-Reviewed Journal Publications

1. **Li, P.**, Chang, C.-Y., and Sadiki, A. (2021). Numerical investigation of pyrotechnical initiators and their impact in airbag inflator applications: Ballistic modeling and functional analysis. *Journal of Energy and Power Technology*, **3**(2), article 026. doi:10.21926/jept.2102026.
2. **Li, P.**, Laudato, M., and Mihaescu, M. (2026). Computational analysis of palatal non-apneic snoring sound generation using a simplified human upper-airway model. *Physics of Fluids*, **38**(8), 081909. **Featured Article**. doi:10.1063/5.0346307.
3. **Li, P.**, Laudato, M., and Mihaescu, M. (2023). Time-dependent fluid-structure interaction simulations of a simplified human soft palate. *Bioengineering*, **10**(11), 1313. doi:10.3390/bioengineering10111313.
4. Wang, B., **Li, P.**, and Li, R. (2016). Symplectic superposition method for new analytic buckling solutions of rectangular thin plates. *International Journal of Mechanical Sciences*, **119**, 432–441. doi:10.1016/j.ijmecsci.2016.11.006.

Selected Unpublished Manuscripts

1. **Li, P.**, Laudato, M., and Mihaescu, M. *Stiffness-dependent modal excitation and force–velocity coupling in fluid-induced soft-palate vibration.* Material-stiffness studies, modal coupling, nonlinear vibration, and acoustic response.
2. **Li, P.**, Laudato, M., and Mihaescu, M. *Effects of tongue-base induced airway constriction on non-apneic palatal snoring.* Geometry-dependent fluid loading, structural response, and pressure/vorticity SPOD.
3. **Li, P.**, Laudato, M., Boij, S., and Mihaescu, M. *Acoustic resonances of a bifurcated duct with a compliant splitter.* Analytical coupled-system modelling, junction impedance, and FEM verification.

Additional Research Experience

Research Project – Non-Isothermal Flow Modelling Aug – Sep 2018
Institute for Fluid Dynamics, Technical University of Darmstadt, Germany
Used COMSOL FEM to investigate Couette flow with a temperature gradient and natural convection in a square cavity, comparing low-Mach-number and Boussinesq formulations.

Research Assistant – Analytical and Numerical Solid Mechanics Jul 2013 – Jun 2015
Institute for Solid Mechanics, Dalian University of Technology, China
Derived analytical plate bending and buckling solutions and compared them with FEM using Mathematica, ABAQUS, and MATLAB.

Scientific Communication

Co-authored *Effects of gravity force and tissue stiffness on a simplified three-dimensional human soft palate*, ICSV30, Amsterdam (2024). Conference participation: EFDC2, Dublin (2025); Swedish Mechanics Days, Gothenburg (2024); Swedish Society of Biomechanics, Lund (2024).

Languages

Chinese – Native | English – IELTS 6.5/9.0 | German – TestDaF 17/20
Swedish – A2 (CEFR)